CSc.111-2075✡

Tribhuvan University

Institute of Science and Technology

2075

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Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 111)
(Digital Logic)
(NEW COURSE)

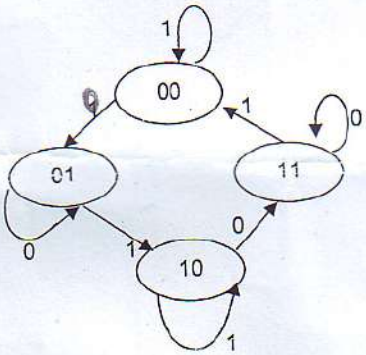
Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
The figures in the margin indicate full marks.

Attempt any two questions: (2×10=20)

1. Implement the following function $F = \Sigma (1, 2, 3, 4, 8)$ using
   (a) Decoder
   (b) Multiplexer
   (c) PLA

2. Design clocked sequential circuit of the following state diagram by using JK flip-flop.



3. The following is a truth a table of a 3-input, 4 output combinational circuit. Tabulate the PAL programming table for the circuit and mark the fuses to be blown in a PAL diagram.

Inputs			Outputs			
X	Y	Z	A	B	C	D
0	0	0	0	1	0	0
0	0	1	1	1	1	1
0	1	0	1	0	1	1
0	1	1	0	1	0	1
1	0	0	1	0	1	0
1	0	1	0	0	0	1
1	1	0	1	1	1	0
1	1	1	0	1	1	1

Attempt any eight questions: (8×5=40)

4. Convert the following decimal numbers to the indicated bases.
   (a) 7562.45 to octal
   (b) 1938.257 to hexadecimal
   (c) 175.175 to binar

5. Express the Boolean function F = A + B' C in a sum of minterms.

6. Reduce the following function using k-map
   F = B'D + A'BC' + AB'C + ABC'.

7. Design a combinational circuit with three inputs, x, y and z, and three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one les than the input.

8. Implement half adder using 2-4 decoder.

9. Design the priority encoder circuit.

10. What is the difference between a serial and parallel transfer? Explain how to convert serial data to parallel and parallel data to serial. What type of register is needed?

11. Design a 4-bit binary ripple counter with D flip-flops.

12. Write short notes on **(any two)**:
   (a) SIMM
   (b) RTL
   (c) Parity Checker